

MedHive: A Proof-of-Concept Dashboard for Preventable Medication Waste Detection in Skilled Nursing Facilities — A Customer Discovery and Feasibility Study

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Abstract

Objective: Preventable prescription medication waste in skilled nursing facilities (SNFs) represents a measurable and underaddressed cost burden on the U.S. healthcare system, with annual losses in long-term care settings estimated at \$2 billion or more. This paper presents MedHive, a proof-of-concept R Shiny dashboard designed to visualize and flag preventable medication waste using Pareto analysis and Statistical Process Control (SPC). **Methods:** A structured customer discovery process was conducted across six SNFs in the Ithaca, NY region via sequential outreach. Three facilities participated in semi-structured informational interviews to assess real-world data availability and validate synthetic data assumptions. **Results:** None of the three interviewed facilities tracked medication-level waste data in a standardized, exportable digital format. Waste documentation was primarily confined to paper Medication Administration Record (MAR) annotations and informal nurse notes. Dashboard architecture was revised to relax schema requirements and support imputed cost fields. **Conclusion:** MedHive demonstrates technical feasibility as a lightweight, open-access waste tracking tool; however, significant gaps between modeled data assumptions and actual facility infrastructure must be addressed before clinical deployment. **Significance:** This work establishes a foundation for scalable medication waste analytics in long-term care settings, with implications for cost reduction, regulatory compliance, and prescriber feedback.

1. Introduction

Prescription medication waste in skilled nursing facilities (SNFs) constitutes a substantial and largely invisible financial burden on the U.S. healthcare system. Bazalo and Weiss estimated that unused solid oral medications dispensed to Medicare Part D residents in nursing facilities cost approximately \$125 million annually, representing 3% of total dispensed cost [1]. More comprehensive estimates that account for all dosage forms and waste sources place the total annual cost of unexpired drug waste in long-term care at approximately \$2 billion, with broader system-wide losses—including hospital and outpatient settings—approaching \$5 billion per year [2]. Early investigations of drug discard patterns in long-term care facilities documented these losses as far back as 1977 [3], yet systematic, facility-level waste tracking tools remain largely absent from clinical practice.

The root causes of SNF medication waste are well characterized in the literature: prescriber discontinuation within days of initiating therapy, patient death or transfer while on long-supply fills, non-adherence, and administrative errors collectively account for the majority of preventable events [4,5]. Despite this knowledge, few of the approximately 15,000 certified SNFs in the United States maintain structured, electronic records of waste at the medication or reason-code level [6]. Electronic health record (EHR) adoption in long-term care lags far behind that of acute care settings, and the facilities that have adopted EHR systems often lack the data granularity or interoperability needed to export waste analytics [7,8].

This paper presents MedHive, a free, open-access R Shiny dashboard designed to address this gap. MedHive accepts user-uploaded CSV data conforming to a defined schema and generates actionable waste insights through Pareto charts, SPC control charts, rule-based operational alerts, and a first-fill policy simulator. The system was developed under a Six Sigma DMAIC

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framework and validated against three synthetic facility datasets representing high-waste, low-waste, and balanced operational profiles. To ground design decisions in operational reality, we conducted a structured customer discovery process targeting SNFs in the Tompkins County, NY region.

The remainder of this paper is organized as follows. Section II reviews relevant literature on medication waste, EHR adoption in SNFs, and healthcare applications of SPC and Six Sigma. Section III describes the customer discovery protocol and dashboard architecture. Section IV presents findings from both the discovery interviews and an illustrative case study using synthetic data. Section V discusses feasibility, limitations, and directions for future work, and Section VI concludes.

2. Literature Review

2.1. Medication Waste in Skilled Nursing Facilities

The problem of prescription medication waste in long-term care is neither new nor trivial. Early observational studies documented drug discard patterns in long-term care facilities as early as the 1970s, with Ciullo and Shepherd reporting that central nervous system agents, anti-infectives, and cardiac drugs collectively accounted for approximately 75% of all Medicaid prescription discards [3]. Brown and Kirk subsequently demonstrated that the introduction of unit-dose drug distribution systems significantly reduced per-patient discard costs in Indiana intermediate-care and SNFs [9], establishing an early precedent for process-level intervention.

Despite these early insights, the scale of the problem grew substantially with the aging of the U.S. population. Morgan estimated that wasted prescription medication among outpatient adults over 65 cost an average of \$30.47 per person per year—modest at the individual level, but projecting to over \$1 billion annually for the national cohort [5]. Bazalo and Weiss subsequently quantified waste specifically within the Medicare Part D SNF population, finding that 6.8% of all dispensed prescriptions were returned partially used and that the cost of unconsumed solid oral medications alone reached approximately \$125 million per year [1]. Lenzer placed these figures in a broader context, reporting that long-term care facilities waste roughly \$2 billion in unexpired medicines annually, with system-wide estimates reaching \$5 billion when outpatient and hospital settings are included [2].

Subsequent work has sought to characterize which patient and medication factors drive preventable waste.

Bekker and colleagues found that over one-third of medications returned to Dutch community pharmacies could be classified as preventable waste, with longer dispensing periods—particularly supplies exceeding three months—carrying significantly elevated risk [4]. Jang and Kang confirmed this pattern in Korean long-term care data, finding that 33.7% of residents had unused medicines and that residents who died during the study period were 15 times more likely to generate waste than survivors [10]. Chambers and Austin highlighted a structural data gap underlying all of these findings: few U.S. long-term care facilities maintain records of the pharmaceutical waste they generate, making system-level estimates uncertain [6].

Responses to this waste burden have ranged from redistribution protocols to formal recycling programs. Toh and Chew demonstrated that 90.8% of donated medications from hospital and clinic sources were reusable, yielding meaningful cost savings [11], while Drummond and colleagues found that recycling unused oral solid medications across a 21-hospital network could divert approximately 461,000 units from incineration with an estimated net value of \$415,000 per year [12]. Smale and colleagues argued that sustainable reduction requires joint responsibility across manufacturers, prescribers, pharmacists, and health authorities rather than any single downstream intervention [13]. Critically, however, all of these downstream interventions presuppose the existence of structured waste data at the facility level—a prerequisite that remains largely unmet in the SNF setting.

2.2. EHR Adoption and Data Infrastructure in Skilled Nursing Facilities

Electronic health record adoption in long-term care settings has historically lagged behind acute care, in large part because the Health Information Technology for Economic and Clinical Health (HITECH) Act explicitly excluded long-term care facilities from its financial incentives [7]. Vest and colleagues found that even among facilities that had adopted health information technology, interoperability was the most frequently cited barrier to effective clinical information sharing, and only facilities classified as “innovative” showed adoption odds more than six times the baseline [8]. Felix and colleagues, examining all federally certified nursing homes in one southern state, found that 69.8% had some form of EHR in use, but that adoption was concentrated among better-resourced facilities operating in competitive markets—suggesting persistent inequities in digital infrastructure [14].

Systematic reviews have catalogued the barriers to

EHR adoption in exhaustive detail. Kruse and colleagues identified 39 distinct barriers across 27 studies, with cost, technical concerns, inadequate support structures, and resistance to change appearing most frequently [15, 16]. Ajami and Arab-Chadegani corroborated these findings, identifying resistance to change as the single most significant implementation obstacle [17]. Gesulga and colleagues further distinguished between people-centered barriers (user resistance and skill deficits) and procedural barriers (concerns about return on investment and insufficient administrative support) [18]. Tsai and colleagues synthesized 142 studies published between 2005 and 2020 and found that EHR implementation produced both positive and negative effects on clinical work, with poor training, resource constraints, and low technology literacy consistently impeding full adoption [19]. Fennelly and colleagues, reviewing national EHR rollouts, identified 15 inter-linked organizational, human, and technological success factors, emphasizing that contextual fit and end-user engagement were as important as technical infrastructure [20].

Where EHRs have been successfully deployed in SNFs, they have demonstrated substantial analytical value. Riester and colleagues used SNF EHR datasets to characterize patterns of *pro re nata* analgesic prescribing and administration across 24,038 hip fracture patients, demonstrating the feasibility of leveraging longitudinal EHR records for pharmacoepidemiology in long-term care [21]. Baughman and colleagues applied structured quality improvement methods to a Veterans Affairs SNF and achieved measurable reductions in unintentional medication discrepancies through comprehensive medication reconciliation protocols [22]. These successes, however, reflect best-case EHR environments; the majority of SNFs—particularly smaller, rural, or under-resourced facilities—lack the data infrastructure to replicate such analyses.

2.3. Six Sigma and Statistical Process Control in Healthcare

Statistical Process Control (SPC) was introduced to healthcare quality improvement in a foundational 2003 paper by Benneyan, Lloyd, and Plsek, who demonstrated that control charts provide a rigorous and accessible method for separating common-cause variation from special-cause signals in clinical process data [23]. Thor and colleagues systematically reviewed 57 empirical studies published between 1990 and 2004 and documented SPC applications across a wide range of healthcare settings, cataloguing 12 categories of benefit and emphasizing that the tool's power depends critically on

correct application and contextual risk adjustment [24].

Subsequent work has extended both the theoretical foundations and the applied scope of SPC in healthcare. Fretheim and Tomic highlighted the underutilized potential of SPC data for rigorous impact evaluation, arguing that time-series data routinely collected in quality improvement programs could support interrupted time series analyses if properly archived and published [25]. Gupta and colleagues demonstrated the application of SPC in complex neonatal intensive care environments, where non-standard data distributions and small sample sizes pose particular methodological challenges [26]. Schmidtke and colleagues reported results from a cluster-randomized trial of SPC training in English NHS hospitals, finding that chart use increased substantially in both intervention and control arms—suggesting a broader cultural diffusion of the methodology across the health system [27].

The Six Sigma DMAIC framework—Define, Measure, Analyze, Improve, Control—provides the organizational scaffold within which SPC tools are most commonly deployed. Monday traced the intellectual lineage of DMAIC from Shewhart's 1920s statistical process theory through Deming's Plan-Do-Study-Act cycle to its modern formalization at Motorola in the 1980s [28]. Ahmed demonstrated that integrating DMAIC with the Theory of Constraints can reduce medical costs, errors, and defects simultaneously by targeting the binding bottleneck within a care process [29], while Reis and colleagues found through systematic review that DMAIC implementations in healthcare consistently reduced length of stay and improved patient satisfaction [30]. Pharmacy applications have proven particularly receptive to Lean Six Sigma methods: Sallam's review of 73 studies found that the most common outcomes were decreased medication turnaround time, improved process efficiency, and reduced medication errors [31]. Thakur and colleagues confirmed these benefits in clinical laboratory settings, where Lean and Six Sigma techniques reduced waste, variation, and work imbalances [32]. A systematic review by de Mendonça and colleagues of SPC applications in abdominal surgery noted that while uptake is growing, methodological quality varies and erroneous conclusions arise when the underlying analytical approach is not rigorously applied [33]—a caution applicable to any new domain of deployment.

2.4. Gap in the Literature

Despite a well-documented epidemiology of medication waste in long-term care, a growing body of SPC methodology, and three decades of EHR adop-

tion research, no lightweight, open-access analytics tool specifically designed for medication-level waste tracking in SNFs currently exists. Prior SPC applications have concentrated in acute care and in-hospital pharmacy settings; EHR literature consistently identifies long-term care as the least-resourced sector for digital data infrastructure; and medication waste research documents the scale of the problem without providing operational tooling for facility administrators. MedHive is designed to occupy precisely this intersection.

3. Methods

3.1. Customer Discovery Protocol

To ground MedHive’s design in operational reality, a structured customer discovery process was conducted targeting SNFs in the Tompkins County, NY region during Fall 2025. Six facilities were identified through CMS Nursing Home Compare and local directories: Kendal at Ithaca, Beechtree Care Center, Cayuga Heights Rehabilitation, the Reconstructive Home, Hospicare of Tompkins County, and Oak Hill Manor. Facilities were contacted via a sequential outreach funnel: (1) cold telephone call or voicemail, (2) follow-up email with a one-paragraph project description, and (3) scheduling of a 15-minute semi-structured informational interview.

Each interview was conducted by telephone and targeted either the Director of Nursing or the facility’s pharmacy liaison. A structured interview guide audited three primary assumptions embedded in MedHive’s initial synthetic dataset: (1) whether facilities tracked medication-level waste data digitally, (2) what variables were routinely captured (e.g., waste reason, drug cost, residual quantity), and (3) what format the data were stored in and whether export was feasible. Interviews were recorded with verbal consent and analyzed thematically for schema validation and design revision.

3.2. Dashboard Architecture

MedHive is implemented as a single-file R Shiny application (app.R) using the packages `tidyverse`, `lubridate`, `scales`, `DT`, `janitor`, and `bslib`. The data ingestion pipeline accepts one or more user-uploaded CSV files conforming to the schema described in Table 1. Each file is processed by the `read_one_csv()` function, which applies column standardization, date parsing, and derivation of three computed fields: per-unit cost (`unit_cost`), waste cost (`waste_cost`), and rea-

son class (`reason_class`), where $\text{unit_cost} = \text{cost_usd} / \text{initial_weight}$ and $\text{waste_cost} = \text{waste_units} \times \text{unit_cost}$.

Table 1. MedHive Minimum Required Data Schema

Variable	Type	Description
prescription_id	string	Unique prescription fill identifier
patient_id	string	De-identified patient code
medication	string	Drug or drug class name
scheduled_date	date	Intended dispense date (YYYY-MM-DD)
refill_date	date	Actual dispense date (YYYY-MM-DD)
cost_usd	numeric	Dispensed cost in USD
prescriber	string	Prescribing clinician name
initial_weight	numeric	Total mass dispensed (grams)
weight_at_stop	numeric	Residual mass at therapy end (grams)*
percent_wasted	numeric	Residual as % of dispensed mass
waste_reason	category	Controlled vocabulary (see text)

*Optional following customer discovery; may be imputed.

Waste reason codes are classified into three operational tiers by the `reason_class()` function: *preventable* (medication change, non-adherence, administrative error, lost medication, expired stock), *semi-preventable* (transfer to hospital), and *non-preventable* (deceased, no waste, other). This taxonomy is used throughout the dashboard to filter views and scope cost-saving estimates.

3.3. Pareto Analysis

For each filtered dataset, MedHive computes total waste cost aggregated by both medication and prescriber dimensions. Items are ranked in descending order of waste cost and a cumulative percentage curve is overlaid, producing a dual-axis Pareto chart consistent with the 80/20 principle used throughout quality management. The Pareto decomposition surfaces the small number of medications or prescribers responsible for a disproportionate share of total waste cost, enabling targeted intervention.

3.4. SPC Control Chart

MedHive aggregates waste cost to the weekly level and applies SPC control limits via the `in_control_limits()` function. Rather than the classical sample standard deviation, the implementation uses a robust estimator based on the Median Absolute Deviation (MAD):

$$\hat{\sigma} = 1.2533 \cdot \text{MAD}(x) \quad (1)$$

where x is the vector of weekly waste costs and the scaling constant $1.2533 \approx 1/\Phi^{-1}(3/4)$ recovers an asymptotically unbiased estimate of the population standard deviation under normality while remaining resistant to outliers [23]. The center line, Upper Control Limit (UCL), and Lower Control Limit (LCL) are then computed as:

$$\text{UCL} = \bar{x} + 3\hat{\sigma}, \quad \text{LCL} = \bar{x} - 3\hat{\sigma} \quad (2)$$

Weekly observations exceeding the UCL are flagged as special-cause variation, warranting investigation by facility staff.

3.5. Operational Alerts

The `flags_tbl()` reactive module generates four categories of rule-based alert. (1) *Policy alerts* flag any medication for which preventable waste constitutes more than 30% of total waste cost and recommend trialing a shorter first-fill supply. (2) *Prescriber alerts* identify clinicians whose preventable waste rate exceeds the facility peer median by more than 10 percentage points, among those with at least 20 attributed prescriptions. (3) *Workflow alerts* flag all waste events linked to patient death where residual medication units are greater than zero, prompting review of auto-halt and EHR notification workflows. (4) *Transfer alerts* summarize waste attributable to hospital transfer events and estimate a 20% recoverable fraction through improved partial-fill and handoff protocols.

3.6. First-Fill Policy Simulator

The `sim_detail()` reactive models projected savings under a partial-dispensing policy. For each preventable waste event i , an initial supply of $q'_i = \max(1, \lfloor w_{\text{init},i} \cdot f \rfloor)$ units is dispensed, where $f \in [0.25, 1.0]$ is the user-specified first-fill fraction. The reducible quantity and units saved are then:

$$\begin{aligned} q'_i &= \max(1, \lfloor w_{\text{init},i} \cdot f \rfloor) \\ s_i &= \min(w_{\text{waste},i}, \max(0, w_{\text{init},i} - q'_i)) \end{aligned} \quad (3)$$

Total projected savings S over the filtered period are:

$$S = \sum_{i=1}^n s_i \cdot c_i \cdot \mathbb{1}[\text{reason_class}_i = \text{preventable}] \quad (4)$$

where c_i is the per-unit cost of prescription i . Results are disaggregated by medication and presented as an interactive ranked bar chart alongside a downloadable detail table.

3.7. Deployment

MedHive is deployed as a stateless, zero-authentication application on `shinyapps.io`, accessible at no cost to any SNF administrator with a web browser. The stateless architecture was selected deliberately for the proof-of-concept phase: it eliminates the need for HIPAA-compliant session management, credential storage, or server-side data persistence, allowing the team to demonstrate core analytical functionality without incurring the security and compliance overhead required for production clinical deployment. The full source code and synthetic demonstration datasets are publicly available at <https://github.com/chatterjee-srinjoy/Control-Freaks-Six-Sigma>.

4. Findings

4.1. Customer Discovery Results

Of the six SNFs contacted, three agreed to participate in informational interviews, yielding a response rate of 50%. The three non-responding facilities either did not return follow-up contact or declined participation citing staff time constraints. Interviews were completed with two Directors of Nursing and one pharmacy liaison across three separate facilities. Table 2 summarizes the facility profiles and key thematic findings.

Several consistent themes emerged across all three interviews. First, none of the facilities tracked medication waste at the individual prescription or reason-code level in any standardized digital format. Waste events were documented informally in paper MAR annotations or nursing shift notes, and this information was not routinely aggregated, costed, or reviewed at an administrative level. Second, all three facilities had some form of return-to-pharmacy workflow for residual medications, but only one could estimate the volume of returned items (by pill count), and none could report aggregate waste costs. Third, the Director of Nursing at Facility F1 identified adverse drug reactions and

Table 2. Customer Discovery Interview Summary (Tompkins County, NY, Fall 2025)

Fac.	EHR System	Role	Key Finding
F1	PointClickCare	Dir. of Nursing	Waste in paper MAR notes; no digital export
F2	MatrixCare	Pharmacy Liaison	Returns tracked by count, not cost or reason code
F3	Paper/Excel hybrid	Dir. of Nursing	Medication lists via weekly fax; no cost attribution

prescriber-initiated medication changes as the most frequently cited informal waste categories, consistent with the controlled vocabulary used in MedHive’s synthetic data. Fourth, all three participants expressed interest in a free, browser-based tool if it could accept data in a format their pharmacy system could generate— though none were confident that their current systems could produce the schema as originally specified.

4.2. Data Availability Gap

Table 3 summarizes the discrepancy between variables assumed in MedHive’s original synthetic dataset and variables confirmed as available through customer discovery interviews.

Table 3. Synthetic Data Assumptions vs. Facility-Reported Availability

Variable	Assumed	Facility Reality
waste_reason	Yes	Informal text; no controlled vocabulary
cost_usd	Yes	Not tracked per-prescription; billed to payer only
weight_at_stop	Yes	Not recorded; return tracked by pill count only
Digital format	Yes	Paper MAR (1 fac.); EHR non-exportable (2 fac.)
Prescriber link.	Yes	Available in EHR; not linked to waste events

4.3. Dashboard Iteration

Based on interview findings, three adjustments were made to MedHive’s data ingestion pipeline and schema requirements. First, `weight_at_stop` was reclassified as an optional field; when absent, the dashboard estimates residual quantity from supply length and standard dispensing unit data. Second, `cost_usd` was made imputable from public drug pricing references (e.g., the Medispan Red Book) using medication name as a join key, allowing facilities without per-prescription cost records to still generate cost-denominated waste estimates. Third, the `waste_reason` controlled vocabulary was supplemented with a free-text annotation field and a simplified three-tier classification (preventable / semi-preventable / non-preventable) as a fallback for facilities unable to provide reason codes.

4.4. Illustrative Case Study

To demonstrate MedHive’s analytical outputs in a realistic operational context, consider a skilled nursing facility comparable to Facility A in the synthetic dataset—a 75-bed SNF with high medication turnover due to frequent transfers and therapeutic changes. Applying MedHive to three months of synthetic data ($n = 302$ prescription events, August–October 2025), the dashboard surfaces the following insights.

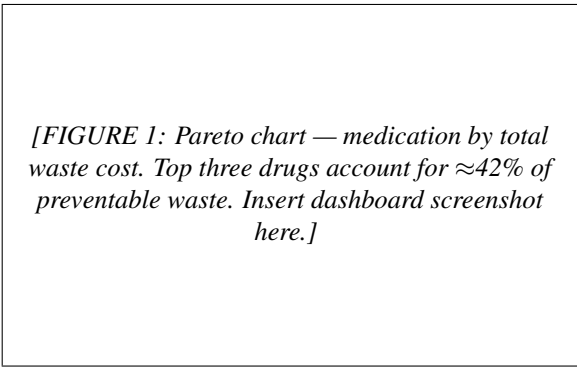


Figure 1. Pareto analysis of medication waste cost, Facility A synthetic dataset ($n = 302$, Aug–Oct 2025). The 80% cumulative threshold is indicated by the dashed reference line.

Pareto analysis (Fig. 1) identifies Risperidone, Warfarin, and Gabapentin as the top three contributors to preventable waste cost, collectively accounting for approximately 42% of total preventable expenditure. All three are high-cost, high-residual medications frequently prescribed on long supply durations—consistent with Bekker et al.’s finding that longer dis-

pensing periods significantly elevate preventable waste risk [4].

[FIGURE 2: SPC time-series chart of weekly waste cost with 3σ control limits. Two out-of-control weeks visible. Insert dashboard screenshot.]

Figure 2. SPC control chart of weekly waste cost, Facility A. The dashed center line represents $\bar{x}$; dotted red lines indicate the UCL and LCL at $\bar{x} \pm 3\hat{\sigma}$ using the MAD-based robust estimator (Eq. 1).

The SPC control chart (Fig. 2) reveals two weeks in which weekly waste cost exceeded the upper control limit, consistent with special-cause variation attributable to a cluster of same-week medication change orders. The use of the MAD-based $\hat{\sigma}$ estimator (Eq. 1) ensures that these UCL/LCL bounds are not artificially widened by the very outlier weeks they are intended to detect.

[FIGURE 3: First-fill policy simulator output at $f = 0.5$. Estimated savings by medication. Insert dashboard screenshot.]

Figure 3. Projected first-fill policy savings by medication, Facility A, at first-fill fraction $f = 0.50$. Only preventable waste events are included in the savings calculation (Eq. 4).

Applying the first-fill simulator at $f = 0.50$ (Fig. 3) projects a 50% partial dispensing policy that, applied to preventable events only, would reduce waste cost by an estimated 28–34% for the highest-impact medications. The operational alert module additionally flags two prescribers whose preventable waste rates exceed the facility peer median by more than 10 percentage points, providing a starting point for targeted prescriber education and feedback.

5. Discussion

5.1. Feasibility Assessment

MedHive’s technical architecture is sound and its analytical outputs are grounded in established quality improvement methodology. The Pareto and SPC tools are standard, well-validated components of the Six Sigma DMAIC framework [23, 24, 31], and the first-fill policy simulator provides a transparent, auditable model for projecting the cost impact of partial-dispensing policies prior to implementation. The dashboard’s stateless, browser-based deployment model minimizes barriers to adoption for facilities that lack dedicated IT infrastructure—a deliberate design choice consistent with the EHR adoption literature’s emphasis on cost and implementation complexity as primary barriers [15, 19].

The central feasibility challenge, however, is not technical but infrastructural: the tool is only as useful as the data that can be supplied to it. Customer discovery revealed that none of the three interviewed facilities possessed the data granularity assumed in MedHive’s original schema. This finding is not surprising in light of the EHR adoption literature [7, 8, 14], which consistently documents that long-term care facilities are the least-resourced sector for structured health data. What the interviews add to this picture is specificity: even facilities using widely adopted commercial SNF EHR platforms (PointClickCare, MatrixCare) do not configure those systems to capture waste reason codes or per-prescription costs at a level exportable to a third-party analytics tool.

The schema relaxations implemented following customer discovery—making `weight_at_stop` optional and enabling cost imputation from public pricing databases—meaningfully lower the barrier to entry. A facility with only a medication name, prescriber, and dispense date can now generate partial analytics; a facility with return-pill-count data can generate Pareto rankings even without unit cost. The minimum viable dataset for meaningful SPC output is weekly prescription volume by medication—a field available from the medication administration records of all three interviewed facilities.

5.2. Limitations

This study has several important limitations. First, customer discovery was limited to three facilities in a single small metropolitan area. Tompkins County, NY is home to a research university and may not be representative of the national SNF landscape in terms of

administrator sophistication, data awareness, or technological infrastructure. Findings should be interpreted as directional rather than generalizable [6]. Second, all performance claims in this paper are based on synthetic data. The three facility datasets were constructed to represent plausible operational profiles (high-waste, low-waste, and balanced), but they have not been validated against actual facility records. Waste cost estimates produced by the policy simulator carry no clinical or financial guarantee. Third, MedHive has not undergone usability testing with SNF staff. The interface was designed by engineers and has not been evaluated by the Directors of Nursing, pharmacy liaisons, or administrators who would be its primary users. Fourth, as a stateless application, MedHive currently stores no data between sessions, preventing longitudinal trend analysis or multi-period benchmarking without re-upload. Fifth, the operational alert thresholds—30% preventable share for policy alerts, 10 percentage points above peer median for prescriber alerts—were selected by the development team based on judgment and the structure of the synthetic data. These parameters have not been empirically validated and should be treated as configurable defaults.

5.3. Future Work

Four directions for future development are prioritized. First, direct EHR API integration with major SNF platforms (PointClickCare, MatrixCare, Yardi Senior Living) would eliminate the CSV upload requirement and allow real-time dashboarding without manual data export [21]. Second, HIPAA-compliant authentication and encrypted session management are prerequisites for any deployment involving real patient data; a Supabase or Firebase backend with role-based access control would address this requirement while preserving the tool's low-cost deployment model. Third, multi-facility benchmarking—allowing a facility to compare its waste metrics against an anonymized peer cohort—would substantially increase the tool's operational value and is a natural extension of the existing multi-file upload capability. Fourth, and most critically, prospective clinical validation in partnership with at least one SNF operator is necessary to establish that MedHive's outputs are interpretable, actionable, and accurate when applied to real facility data. Such a study would simultaneously validate the revised schema, test the operational alert thresholds, and provide the first empirical cost-impact estimates outside of simulation.

6. Conclusions

This paper has presented MedHive, a proof-of-concept R Shiny dashboard for preventable medication waste detection in skilled nursing facilities. Developed under a Six Sigma DMAIC framework, MedHive integrates Pareto analysis, MAD-based SPC control charts, rule-based operational alerting, and a first-fill policy simulator into a unified, zero-cost, browser-accessible interface.

A structured customer discovery process conducted across six SNFs in Tompkins County, NY found that none of the three participating facilities tracked medication waste at the individual prescription or reason-code level in an exportable digital format—confirming that the primary barrier to adoption is not tool availability but data infrastructure. In response, the dashboard schema was revised to support optional and imputed fields, substantially lowering the minimum data requirements for useful output.

Illustrative case study analysis using synthetic facility data demonstrated that MedHive can surface actionable insights: Pareto decomposition identifies the small number of medications driving disproportionate waste cost; SPC control charts detect anomalous waste weeks not attributable to routine variation; and the policy simulator projects meaningful savings from even modest first-fill fraction reductions.

MedHive is freely available at <https://github.com/chatterjee-srinjoy/Control-Freaks-Six-Sigma> as an open-access tool for the SNF community. The authors hope that this proof-of-concept will stimulate further investment in medication waste analytics infrastructure for long-term care, and that future work will deliver a clinically validated, HIPAA-compliant version of the dashboard in partnership with operating facilities.

Acknowledgment

The authors thank Dr. Timothy Fraser and the Cornell Department of Systems Engineering for guidance throughout this project. This work was conducted as part of the Cornell Meinig School of Biomedical Engineering M.Eng. program. The authors also thank the nursing home administrators and Directors of Nursing in Tompkins County, NY who generously participated in informational interviews.

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A. Appendix: Minimum CSV Schema and Source Code

The MedHive source code, synthetic demonstration datasets (Facilities A, B, and C), and the codebook documenting all schema variables are available at:

```
https://github.com/chatterjee-srinjoy/Control-Freaks-Six-Sigma
```

The minimum required CSV fields for SPC output are: `prescription_id`, `medication`, `scheduled_date`, and `waste_reason`. Full Pareto and cost-denominated outputs additionally require `cost_usd` (or an imputed cost via medication name lookup) and `initial_weight`. The policy simulator requires `initial_weight`, `weight_at_stop` (or an estimate), and `cost_usd`.